

Simulation of Fabrication and Degradation of All-Solid-State Batteries with Ductile Particles

To cite this article: Magnus So *et al* 2021 *J. Electrochem. Soc.* **168** 030538

View the [article online](#) for updates and enhancements.

You may also like

- [Multi-Physics Simulation of Solid-State Batteries with Active Material Coating](#)
Baber Javed and Michihisa Koyama
- [Understanding the Effect of Mechanical Degradation on the Performance of Solid-State Batteries through Particle Simulations](#)
Magnus So, Agnesia Permatasari, Shinichiro Yano et al.
- [Quantifying Resistive and Diffusive Kinetic Limitations of Thiophosphate Composite Cathodes in All-Solid-State Batteries](#)
S. Yanev, H. Auer, R. Pertsch et al.

Your Lab in a Box!

The PAT-Tester-i-16 Multi-Channel Potentiostat for Battery Material Testing!

- ✓ **All-in-One Solution with Integrated Temperature Chamber (+10 to +80 °C)!**
No additional devices are required to measure at a stable ambient temperature.
- ✓ **Fully Featured Multi-Channel Potentiostat / Galvanostat / EIS!**
Up to 16 independent battery test channels, no multiplexing.
- ✓ **Ideally Suited for High-Precision Coulometry!**
Measure with excellent accuracy and signal-to-noise ratio.
- ✓ **Small Footprint, Easy to Setup and Operate!**
Cableless connection of 3-electrode battery test cells. Powerful EL-Software included.



EL-CELL®
electrochemical test equipment

Learn more on our product website:



Download the data sheet (PDF):



Or contact us directly:

☎ +49 40 79012-734

✉ sales@el-cell.com

🌐 www.el-cell.com



Simulation of Fabrication and Degradation of All-Solid-State Batteries with Ductile Particles

Magnus So,¹ Gen Inoue,² Ryusei Hirate,³ Keita Nunoshita, Shota Ishikawa, and Yoshifumi Tsuge

Department of Chemical Engineering, Faculty of Engineering, Kyushu University, Nishi-ku, Fukuoka 819-0395, Japan

We simulated cold press fabrication and intercalation damage in a sulfide All-Solid-State Battery (ASSB) electrode using the Discrete Element Method. We developed a new cohesive hybrid-particulate model that both can simulate particle consolidation during fabrication and material failure during intercalation expansion. In this way, the effect of the fabrication conditions on the mechanical degradation of the electrode can be simulated. The high pressure in the cold press fabrication cause plastic deformation and build-up of cohesive contacts between the particles, consisting of Si active material (AM) and sulfide solid electrolyte (SE), resulting in densification of the electrode. During charging, when AM expands during lithiation, the AM-SE contact area increases but the effective SE conductivity decreases. When the expansion is small, the contact area and conductivity may recover to their original value. However, large expansion may cause plastic deformation and cracking that cause permanent reduction of both contact area and SE conductivity. This type of mechanical degradation was significantly less for electrodes fabricated at higher pressures. This model can become a valuable tool to improve the durability and performance of future ASSBs.

© 2021 The Electrochemical Society ("ECS"). Published on behalf of ECS by IOP Publishing Limited. [DOI: [10.1149/1945-7111/abcd23](https://doi.org/10.1149/1945-7111/abcd23)] All rights, including for text and data mining, AI training, and similar technologies, are reserved.

Manuscript submitted December 17, 2020; revised manuscript received February 10, 2021. Published March 26, 2021. *This was paper 998 presented during PRiME 2020, October 4–9, 2020.*

All-Solid-State Batteries

For electric vehicles and energy storage applications, there are requirements of higher performing, durable and safer batteries than current Li-ion batteries. In All-Solid-State batteries (ASSBs), the liquid electrolyte is replaced by solid electrolyte (SE). The SE is inflammable which significantly eliminates the risk of fires.¹ The SE also plays a role as a separator and the stack can be made more compact which increases the power density.

In a review, Sun et al.² identified the following issues and challenges regarding ASSBs: (i) poor interfacial combability between AM and SE, (ii) insufficient ionic conductivity, (iii) insufficient understanding of interfacial processes and (iv) insufficient system design and optimization.

High capacity AM materials such as Si have developed with a capacity of ~4200 mAh g⁻¹.³ However, the material may expand up to nearly 300% which may cause severe mechanical degradation.^{4–6} Expansion cause cracks within the SE material which inhibit Li + conduction through the SE. Contraction may cause delamination of the AM-SE interface and buildup of shell voids around AM which reduce AM-SE interface contact.

Recently, many new promising SE materials such as anti-perovskite, argyrodite, garnet, Li₃N, LiPON, LISICON, NASICON, perovskite and sulfide have been developed. Not only have they ion conductivities comparable to liquid electrolyte but also low electronic conductivities and wide electrochemical windows.⁷ However, their full potential cannot be utilized if there is any crack formation or delamination of the AM-SE interface, so several studies have focused on SE mechanical properties. SE mechanical properties are important for mitigation of lithium dendrite growth, reducing AM-SE interfacial resistances and avoiding crack formation.⁸ Glassy sulfides such as Li₂S have shown many of the favorable mechanical properties. The Young modulus of elasticity is low which can reduce stresses during lithiation expansion. Sulfides can also be pressure-sintered in room temperature which eliminates problem with secondary reactions at high temperature that would otherwise reduce the performance.⁹ Sulfide SE are also ductile with a Pugh's ratio of ~2.4 which has been shown to be directly related to their ability to form a dense electrode.¹⁰

Reconstructions models of ASSB.—The reconstruction models of ASSB found in literature range from nanoscale up to the

mesoscale. In this study we focus on the microscale for the reconstruction of the ASSB electrode. The microscale reconstruction models can be subdivided into top-down and bottom-up approaches. In top-down approaches the models are derived from the larger scales of the system, whereas the reverse is conducted in bottom-up models

Top-down microscale reconstruction models of ASSB.—One of the simplest form of top-down microscale model is the one-dimensional model developed by Fabre et al.¹¹ Recent experimental observation methods and advancement of computer power has enabled the development of 3D reconstruction top-down reconstruction models. Inoue and Kawase¹² developed a three-dimensional top-down reconstruction of Li-ion battery electrode including graphite and lithium cobalt oxide using focused ion beam-scanning electron microscopy (FIB-SEM). They found that the particle morphology and binder distribution have a large impact on the mass transfer capability. By performing electrochemical reaction and mass transfer in ASSB, Itoh et al.¹³ simulated a three-dimensional interdigitated structure and compared it with an electrode with a laminated structure. The interdigitated structure resulted in a better performance due to reduced diffusive length of Li-ion.

Fathiannasab et al.¹⁴ obtained the ASSB electrode structure using a synchrotron transmission X-ray microscopy tomography system and the galvanostatic discharge was simulated. The void spaces hindered the ionic mass transfer especially at high current rates. Electrodes fabricated under a higher pressure had lower porosity and more visible void spaces.

Bottom-up microscale reconstructions models.—The simplest bottom-up models use static particles with positions calculated with a distribution algorithm without consideration of particle forces. Bielefeld et al.¹⁵ used a particle based model to evaluate connectivity and AM-SE interface contact area with different porosities and AM/SE compositions. The model was in a later study extended to calculate tortuosity with different particle size distributions.¹⁶

DEM is a particle-based method to solve the interaction in granular materials.¹⁷ There are few DEM models that focus on ASSB. Gimenez utilized a DEM packing model to estimate the electronic conductivity of AM and CB particles.¹⁸ The model implemented only small deformations of particles and assigned the void space between AM particles as the SE material. Shi et al.¹⁹ used DEM to perform a particle size optimization of AM and SE and

²E-mail: ginoue@chem-eng.kyushu-u.ac.jp

concluded that smaller sizes of SE in contrast to AM was beneficial for mass transfer.

DEM plastic models for cold pressing application.—There are several DEM plasticity models that have been used for cold-pressing applications outside the context of ASSB. Plasticity models for DEM can be steady, such as Storåkers model,²⁰ which do not include spring-back during unloading conditions. However, in order to model the finished electrode, we need a model that also include the permanent deformation. In this way we can model the transition from the granular material in early fabrication process to the solid material of the final electrode product. DEM in its basic form is ideal for modelling granular materials. However, there are also hybrid-particulate models of DEM that can simulate materials with the use of cohesive beams which may break upon failure.²¹ In these cases, the strength of the cohesive beams is related to the tensile strength of the material.

In molds on the other hand, build-up of cohesive connections is possible through application of high pressure or temperature through either cold or hot pressing. Two examples of models that include such effects can be found in the literature by Tomas²² and Luding.²³ In these models it is also possible to simulate spring-back conditions during unloading. The DEM model developed in this study was based on these studies.

Modelling failure of battery electrode.—Bucci et al.²⁴ used a FEM model to simulate fracture in a ASSB electrode. The AM was modelled as expanding cubes and SE was modelled as the continuous phase. The effect of particle inhomogeneity was incorporated into the parameters of the model. Fracture of SE was modelled by the use of cohesive theory of fracture which incorporates a cohesive zone around the cracks that limits the cracks progression. The fracture of the SE material occurs if AM is allowed to expand more than 7%.

In a later study, Bucci et al.²⁵ developed a one-dimensional analytical model of the AM-SE interface based on the cohesive theory of fracture. The delamination criterium and a stability condition for fracture propagation on the AM-SE interface was developed. Material property mapping was conducted to identify regimes where failure was initiated by either fracture or plastic deformation. It was concluded that plasticity of the electrode materials could significantly prevent the onset of delamination between AM and SE.

There are few studies that have incorporated plastic deformation of ASSB electrode. Di Leo et al.²⁶ developed a model of plastic deformation of Si nanotube anodes for Li batteries. The nanotubes were covered by a stiff silicon-dioxide shell which prevented outward expansion and Si deformed plastically towards the inside of the nanotubes. When the stress exceeds the yield stress, plastic flow occurred which caused the nanotubes to deform. Gao et al.²⁷ simulated contact stresses of Si-C core-shell anode particles for Li-ion batteries. During intercalation expansion the particles deformed plastically. Liu et al.²⁸ developed a stress contact model of the solid

electrolyte interface (SEI) layer surrounding AM particles in an ASSB electrode. Increased plastic deformation of SEI caused it to wrinkle and increased the risk of ratcheting and failure during cyclic voltage

Purpose and scope of the study.—Ductility is essential to form a compact ASSB electrode and an intimate contact with AM.^{9,10} Calculations have also shown that ductility can reduce internal stresses during intercalation expansion.²⁵ Despite of the importance of the ductility of the SE materials there are, to our best knowledge, currently no model that can simulate its effect on cold press densification and intercalation damage.

We will demonstrate a two-step simulation including: cold press fabrication and cyclic operation. In this way we can model how fabrication conditions affect the intercalation damage. The study involves the development of a new DEM plastic deformation model for ASSB electrode. During fabrication simulation, the model can simulate build-up of cohesive contacts when the particles are subject to high compressive forces. The model was calibrated by simulating with different fabrication pressures and comparison with previous experimental results. During cyclic operation the active material expands periodically which cause stresses and eventually failure and fracture within the electrode. With the new model it is possible to model the effect of fabrication conditions on the durability and performance of the electrode.

Methods

Discrete element method.—An overview of the particle-particle interaction of a common DEM model is shown in Fig. 1. As depicted in Fig. 1a, spring and damper models are coupled to include the effect of viscous damping and elasticity between particles in the normal and tangential components. When two particles are in contact, they deform and exert force on each other. The overlap distance, h_{ov} , [m] depicted in Fig. 1b, is a measure of this deformation. The parabolic stress distribution of two spheres is also shown with a contact radius of $a = \sqrt{h_{ov} R_{eff}}$ [m]. The Hertzian contact law express the normal spring force, F_{normal} [N], as follows:

$$F_{normal} = k_{spring} h_{ov} \quad [1]$$

where k_{spring} [N m⁻¹] is the spring coefficient, given by

$$k_{spring} = 4/3 \cdot E_{eff} \cdot h_{ov}^{1/2} \cdot R_{eff}^{1/2} \quad [2]$$

where R_{eff} [m] is the effective radius based on the radii of the interacting particles, R_1, R_2 [m]

$$R_{eff}^{-1} = R_1^{-1} + R_2^{-1} \quad [3]$$

and E_{eff} [Pa] is the effective elastic modulus given as a function of the Young moduli, E_1, E_2 [Pa] and the Poisson ratios, ν_1, ν_2 [—], as follows:

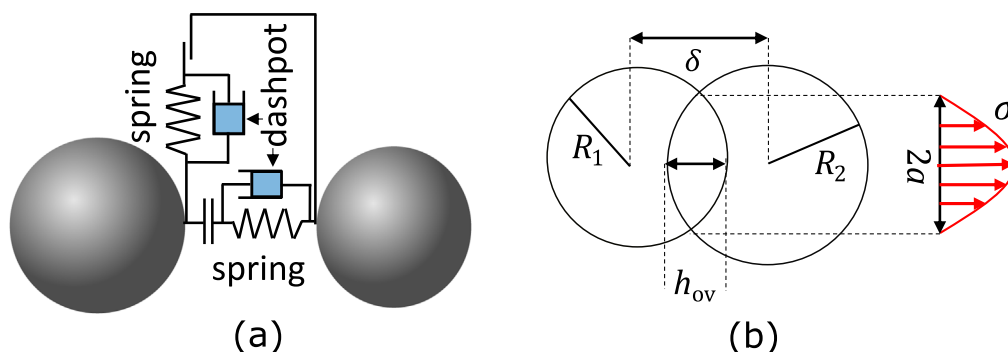


Figure 1. Overview of contact force interaction in DEM: (a) spring and dashpot interactions, (b) overlap distance and stress distribution.

$$E_{\text{eff}}^{-1} = (1 - \nu_1^2)E_1^{-1} + (1 - \nu_2^2)E_2^{-1}. \quad [4]$$

Apart from the normal elastic force, we also include damping and slide friction forces and detailed description of these interactions can be found in the original DEM model.¹⁷

New model of plastic deformation.—A new plasticity model was developed and was adopted from Tomas²² and Luding.²³ As with Tomas,²² it follows the elastic Hertzian contact law under low stress and permanent plasticity deformation occur when the shear stress exceed the hardness of the material. As with Luding,²³ the interactions in our model are simplified but differs from Luding²³ as the spring coefficient is not linear but based on the Hertzian contact law. The linear model of Luding²³ assume that plastic deformation occurs almost immediately which is applicable for materials of low hardness vs elasticity by the relation: $\sigma_H/E < 0.01$ which is not the case for all solid electrolytes.

The differences between elastic and plastic deformation are shown by comparing Fig. 2a with Fig. 2b, where a permanent deformation can be described by the change of the equilibrium overlap. With the equilibrium overlap, it is possible to simulate the spring-back effect after unloading. In case of pure elastic deformation, the equilibrium overlap is zero and the force is determined solely by the overlap distance and the deformation is reversible. In case of plastic deformation, there is a permanent deformation which can be described by the equilibrium overlap in Fig. 2b. The normal force is the spring coefficient times the difference between the overlap and equilibrium overlap

$$F_{\text{normal}} = k_{\text{spring}}(h_{\text{ov}} - h_{\text{eq}}) \quad [5]$$

and because we use Eq. 2 for the spring coefficient, our model follows the Hertzian contact law if h_{eq} is zero. We simulate the plastic deformation by the rate of change of the equilibrium overlap

with a relaxation time, t_{rel} , as follows:

$$\frac{\partial h_{\text{eq}}}{\partial t} = \frac{(F_{\text{normal}} - F_{\text{threshold}})}{t_{\text{rel}} k_{\text{spring}}} (1 - \phi_{\text{solid}}^4) \quad [6]$$

The parameter t_{rel} can be viewed as a time scale for plastic flow. The time scale for real plastic flow and the time for pressing are several orders of magnitudes greater than the simulation time step. Therefore, using the real time scales would lead to unacceptably long simulations. Because we are interested in the steady state solution, we set these timescales arbitrary and ensured that t_{rel} was much smaller than the simulation time.

The plastic flow will be triggered above the threshold force which can be calculated from the hardness of the material. The maximum stress σ_{max} is a factor 3/2 higher than average and is given by:

$$\sigma_{\text{max}} = 3/2 \cdot F_{\text{normal}}/A_{\text{con}} \quad [7]$$

If the hardness and contact area are known, we can estimate the threshold force as follows:

$$F_{\text{th}} = 2/3 \cdot \sigma_H A_{\text{con}} \quad [8]$$

For smooth surfaces A_{con} can be easily obtained. However, the surface of particles in ASSB electrode is not completely smooth.²⁹ Although there are models to account for roughness,³⁰ we use a single parameter α which is the ratio of the real contact area in comparison with a sphere with smooth surface. The contact area can thus be estimated as follows with α as a fitting parameter:

$$A_{\text{con}} = \pi h_{\text{ov}} R_{\text{eff}} \alpha \quad [9]$$

The plastic deformation will stop once the force is less than the threshold. Strain hardening is often assigned to plasticity models; however, in this model it was neglected due to lack of experimental data. Under plastic deformation with a constant load, the stress will

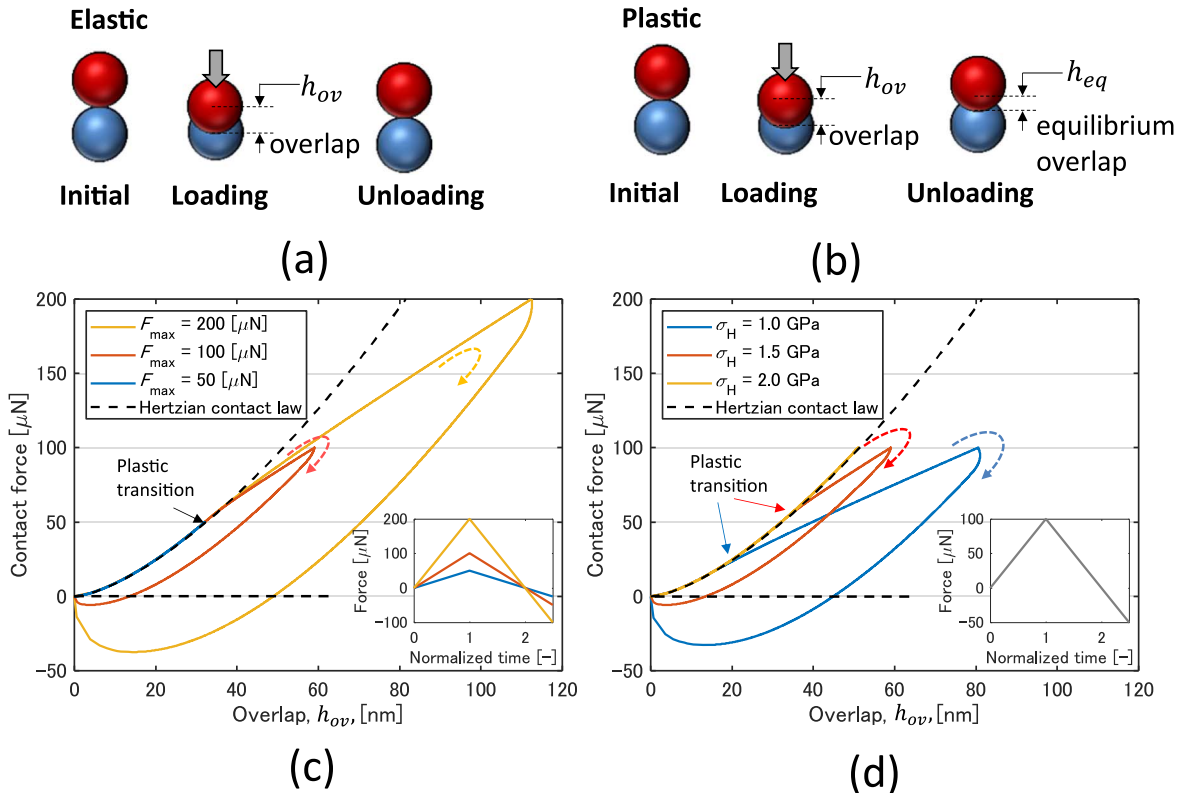


Figure 2. Overview of contact force interaction in our DEM model for elastic (a) and plastic (b) deformation. Force displacement hysteresis curves with different forces (c) and different hardness (d).

decrease with time, due to an increase of the contact area, until the stress no longer exceeds the hardness.

In order to avoid over-compaction, we inserted a term in the end of Eq. 4 that goes to zero when the estimated the equilibrium solid volume fraction goes to unity. This means that the local volume fractions of the particles needed to be estimated. Although our simulations were conducted in 2D, for the purpose of volume fraction estimation, we modelled with a domain that had thickness in the z-direction equal to the average SE diameter. We estimated the local volume fraction by first subdividing the volume around the particles by polygonal prisms. The polygonal sides of the prisms were estimated by a Voronoi tessellation of the particle positions. Finally, the local solid volume fractions were obtained by dividing the volume of the particles with the volume of the prisms.

The analysis of the cold pressing of sulfide electrolyte by Sakuda et al.⁹ indicates that atoms and/or ions diffuse at the material/pore interface, forming strong bonds between particles during densification. Therefore, we assumed that sulfide electrolyte particles, that have undergone plastic deformation upon contact, will exert tensile stresses during separation.

During lithiation, tensile stresses may cause crack growth in the SE material and at the AM-SE interface. The tensile failure modes of in ASSB are initiated by either plastic failure or fracture by the cohesive zone theory.²⁵ We added a condition that if the tensile stress exceeds the hardness of SE, the particles lose their connection. In order to for this assumption to be valid, the hardness related to plastic deformation σ_H has to be lower than the fracture strength. In the cohesive theory of fracture, the fracture strength σ_c can be estimated by

$$\sigma_{coh} = 2\gamma/\delta \quad [10]$$

where γ is the cohesive energy of the crack and δ is the length of the cohesive zone. For Li_2S electrolytes, γ has been estimated to $2.8 \pm 1.8 \text{ J m}^{-2}$,^{24,31} and the values of δ can be roughly estimated to 1 nm.²⁵ Hence, Eq. 7 results in a fracture strength of $\sigma_c = 5.6 \text{ GPa}$ and a hardness to fracture strength ratio (σ_H/σ_c) of 0.34. The initial failure of Li_2S can therefore be assumed to be dominated by plastic deformation.

In order to graphically illustrate our model, we use force-displacement curves similar to indentation test in Figs. 2c and 2d. The force F increase linearly until maximum force, F_{max} , and then decrease linearly. The effect of F_{max} on the hysteresis curve can be seen in Fig. 2c. Here the force is plotted with h_{ov} . Hysteresis can be seen in the force-overlap curve when the force exceeded the threshold for plastic transition. For the red and yellow lines, the new equilibrium overlap has moved to the position where the force curve crosses the x-axis during unloading. The effect of the material hardness can be seen in Fig. 2d. The material hardness will affect the force where plastic transition occurs and also the final equilibrium overlap.

Implementation of DEM to simulate cold pressing and intercalation damage of ASSB electrode.—In the first set of simulations, we simulated only solid electrolyte for comparison with experiments of Sakuda et al.⁹ The SE material was a sulfide type, specifically: lithium thiosphosphate glass with 75% dilithium sulfide. The boundary conditions were wall boundary on the bottom, pressure wall boundary at the top and periodic boundary conditions on the sides. The particles were grouped in clusters with diameter of 5–10 microns that corresponds to SEM observations of Sakuda et al.⁹ The initial particle positions were calculated by pre-existing algorithms, originally applied for fuel cells.^{32,33} Firstly, the aggregation of the particles into clusters was calculated. Finally, the particle positions were determined by performing a DEM simulation with gravity.

In the next step, we simulate the anode electrode including AM particles. These simulations are divided in two parts: cold pressing and intercalation. The boundary conditions in the former were the same as the calibration simulation. In the intercalation simulation, the AM particles expand due to lithiation. On the top wall we added a spring force which resemble the compression of nearby components. The spring force was calculated by assumption that volume expansion will compress the SE layer and the cathode layer. The areal spring constant was estimated by dividing the SE Young modulus E_{SE} with the sum of the SE layer thickness, L_{SE} , and the cathode thickness, L_{cat} . We developed our DEM model in MATLAB[®] and ran the calculations on a conventual PC. A list of parameters used in the simulations can be seen in Table I. The cold pressing simulations were conducted with different fabrication pressures, P_{fab} . The Si intercalation expansion ratio β_{Si} quantifies the relative expansion of a fully charged AM particle. The State of Charge (SOC) is defined as the instantaneous charge divided by the charge capacity of the material. Because the expansion of Si is relatively large, we only partially intercalate the Si material and the maximum value was set to SOC_{max} . The coefficient of restitution, e , determine the bounciness of the particles and the friction coefficient μ determine the sliding friction resistance between the particles. The parameter ϕ_{AM}^{rel} is the AM volume fraction relative to all solids in the anode.

Results

Simulation of SE cold pressing for calibration.—Time series of the fabrication pressure can be seen in Fig. 3a. The pressure increases first rapidly with time and eventually relaxes at the fabrication pressure. Time series of the height can be seen in Fig. 3b. The height of the electrode layer reduces exponentially with time until the pressure releases. During unloading the height increases but do not go all way back to the initial position. We can see that the higher the fabrication pressure, the lower the final height. The initial positions of the particle clusters can be seen in Fig. 3c. The final position of the SE particles after unloading can be

Table I. List of parameters used in the simulations.

Parameter	Value	Description	Reference
e	0.5	Coefficient of restitution	This study
E_{SE}	24 GPa	Young modulus of sulfide SE	Sakuda et al. ⁹
E_{AM}	70 GPa	Young modulus of Si AM	Sethuraman et al. ³⁴
L_{cat}	60 μm	Thickness of cathode layer	This study
L_{SE}	60 μm	Thickness of SE layer	This study
P_{fab}	50–500 MPa	Fabrication pressure	This study
SOC_{max}	8%–24%	Maximum State of Charge	This study
μ	0.5	Friction coefficient	This study
ν	0.3	Poisson ratio	This study
β_{Si}	2.8	Si expansion factor	Obrovac et al. ³⁵
σ_H^{SE}	1.9 GPa	Hardness of SE	McGrogan et al. ³¹
ϕ_{AM}^{rel}	0.4	AM relative volume fraction	This study

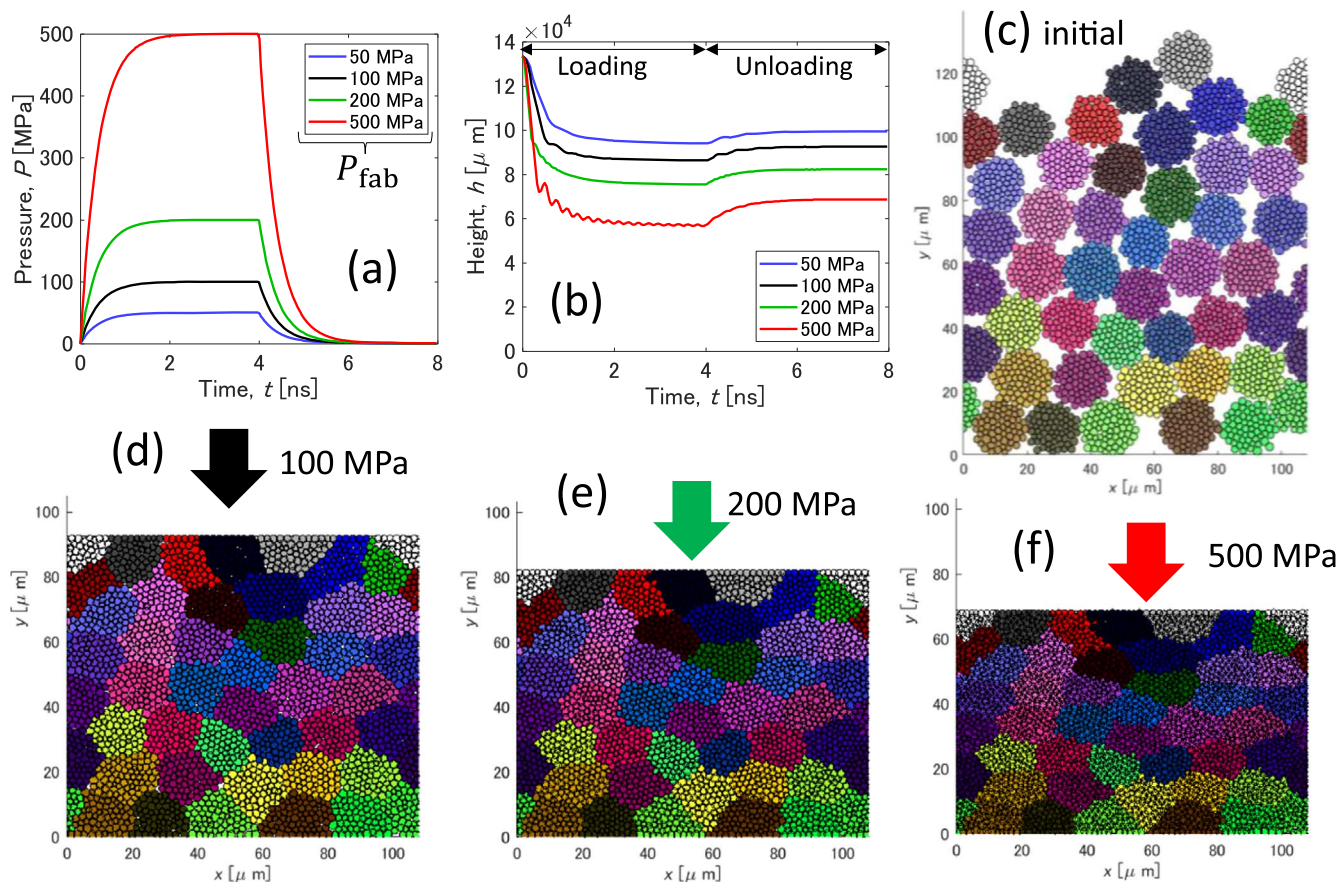


Figure 3. Time series of fabrication pressure (a) and height. Initial SE particle cluster before cold pressing (c) and after pressing (d)-(f).

seen for 100 MPa (d), 200 MPa (e) and 500 MPa (f). The results in Fig. 3 were obtained with $\alpha = 0.65$.

Figure 4 compares our simulations with experimental results of Sakuda et al. (2013)⁹ and the relative density is plotted with the fabrication pressure in (a). We used an indentation hardness of 1.9 GPa³¹ from literature with a similar Li_2S sulfide content as

Sakuda et al. (70% instead of 75%). The colored lines are from our simulations and the black rings represent cold pressing experiments from Sakuda et al.⁹ The relative SE density was measured by Sakuda et al.⁹ by comparing the density of cold pressed pellets with hot pressed pellets. We can see that for the simulation with a relative area of $\alpha = 0.65$, the results of relative density agree well with

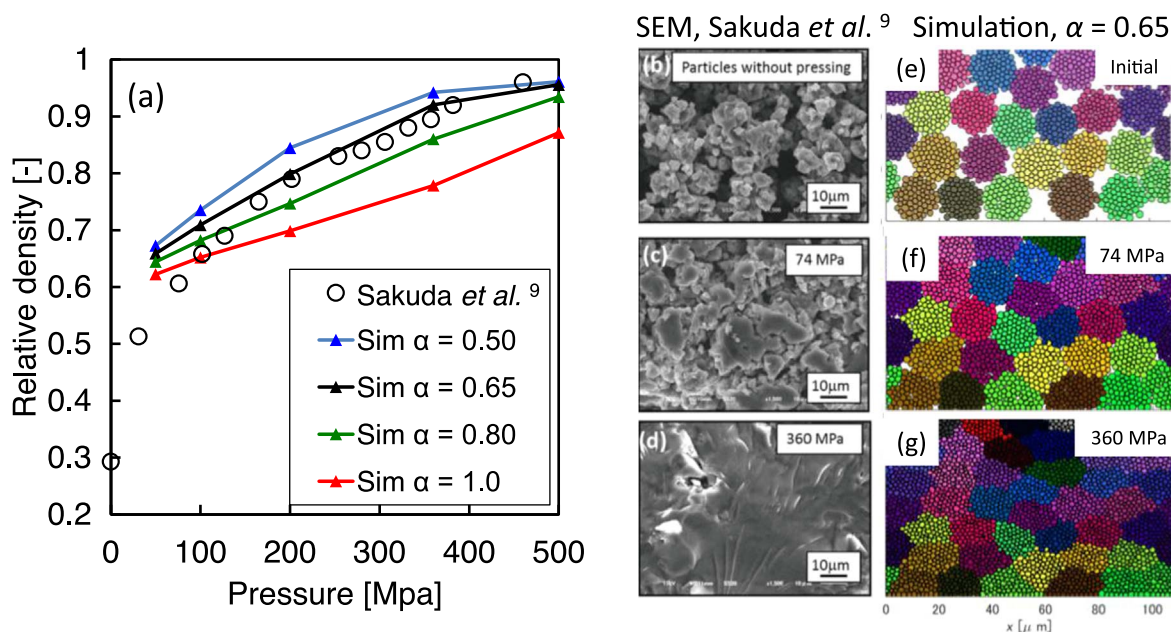


Figure 4. The relative density of SE plotted with the fabrication pressure for the simulations and experiments.⁹ Comparison of local particle structures from SEM images (b)-(d) and simulation (e)-(g). SEM images (b)-(d) reproduced from Sakuda et al.⁹ by open access permission from Springer Nature.

experiments for higher pressures 200–500 MPa, but the differences are significant at lower pressures. Figures 4b–4d are Scanning Electron Microscope (SEM) images from Sakuda et al. (2013).⁹ Voids are present in the case of low pressure (b and c) but collapses at higher pressure (d). Corresponding structures obtained from simulations are shown in Figs. 4e–4g for $\alpha = 0.65$. The difference of the relative density between simulation and experiment at lower fabrication pressures can be explained by comparing (b) with (e). The real aggregate structure (b) causes larger voids because of its 3D aggregate structure.

Simulation of cold pressing of ASSB electrode.—Time series of the electrode height can be seen in Fig. 5a. As with the previous simulation of SE, the height first decreases exponentially with time and stabilizes and increase during the unloading stage. The particle distribution can be compared from the initial distribution (b) with the final (c–e) with increasing fabrication pressure.

Simulation of intercalation of ASSB electrode.—*Stress distribution.*—Figures 6a, 6b show the distribution of the stress in SE during lithiation with different SOC. As expected, the stresses increase with the SOC value. A histogram of the stress distribution can be seen in Fig. 6c. Positive values indicate compressive stress and negative values tensile stress. We can see that a major portion of the shear stress is higher than the hardness of the SE material (1.9 GPa³¹).

Intercalation damage.—Figure 7 show the effect of the intercalation expansion on the electrode after 5 cycles. The subfigures in the top row show the results for an electrode fabricated at 100 MPa. The results of SOC 8% are relatively unaffected, but cracks can be seen for SOC 24%. The bottom subfigures show the results for an

electrode fabricated at 200 MPa. The electrodes are significantly less damaged compared to those fabricated at 100 MPa, especially for the case of SOC 24%. The much denser electrode from the high fabrication pressure show much less cracking damage after intercalation expansion.

Time series of intercalation pressure, electrode height and porosity.—The time series of the wall pressure during intercalation is shown in Fig. 8a. When AM expands it will periodically cause a high pressure on against the SE layer. The time series of electrode height can be seen in Fig. 8b. Time series of porosity can be seen in Fig. 8c and shows that porosity increases during intercalation expansion and increases with SOC_{max}. As plotted in Fig. 8d, higher fabrication pressure causes lower overall porosity.

Tortuosity and SE conductivity.—Time series of tortuosity in SE, τ_{SE} , can be seen in Figs. 9a and 9b. The tortuosity was calculated by projecting the SE particles on 100 images per time series. The tortuosity of each image was calculated using TauFactor which is an open source MATLAB application.³⁶ The time series of τ_{SE} are plotted with different SOC_{max} (a) and fabrication pressure (b). Time series of the relative ionic conduction coefficient, κ_{SE}^{rel} , is shown in Figs. 9c and 9d. It is defined as the ratio of the effective conductivity κ_{SE}^{eff} to the bulk value κ_{SE}^{bulk} and can be obtained from the ratio of SE volume fraction, ϕ_{SE} , and τ_{SE} as follows:

$$\kappa_{SE}^{rel} = \frac{\kappa_{SE}^{eff}}{\kappa_{SE}^{bulk}} = \frac{\phi_{SE}}{\tau_{SE}} \quad [11]$$

From the periodic shapes of the curves, we can see that κ_{SE}^{rel} is higher during delithiation when SE particles are more closely packed. At

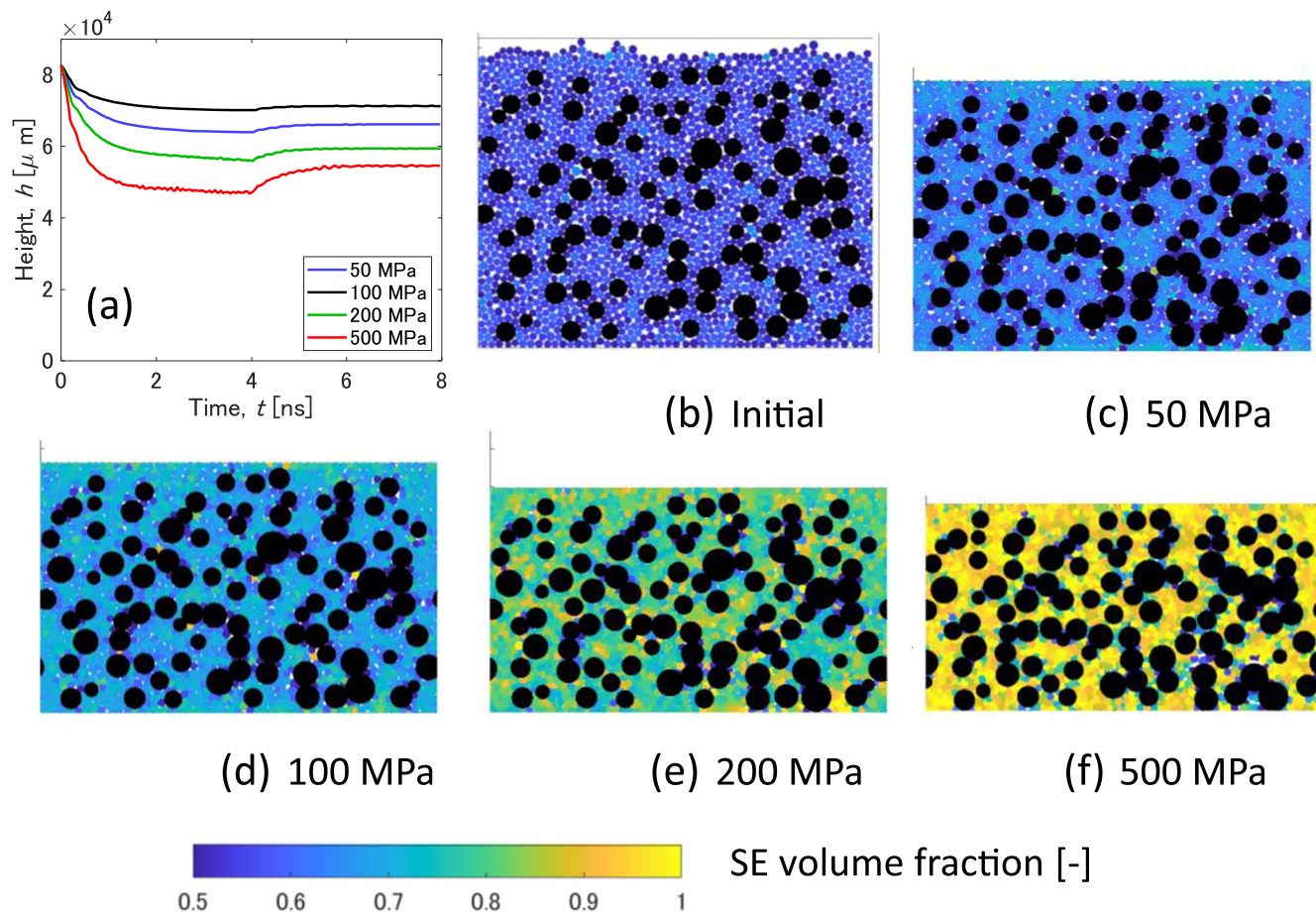


Figure 5. Time series of electrode height (b). Electrode particles before cold pressing (c) and after pressing (d–f).

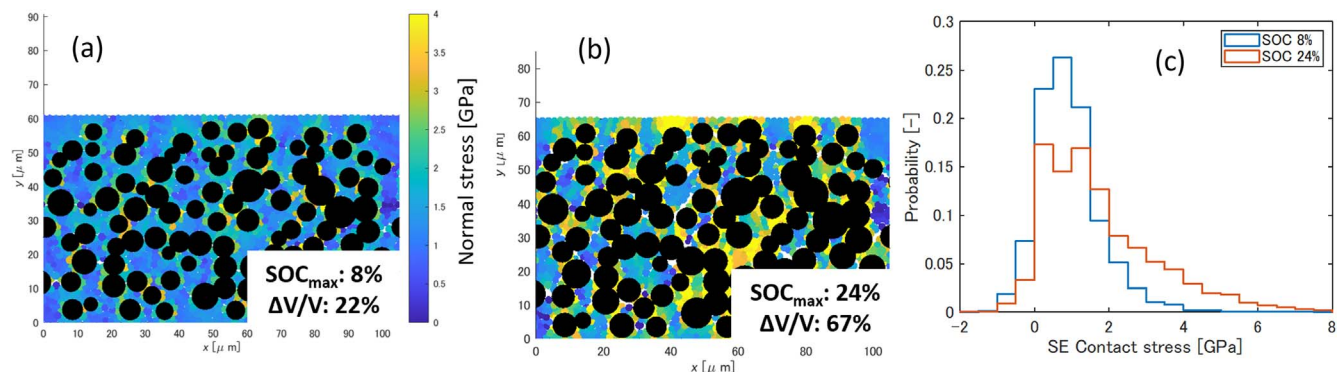


Figure 6. Distribution of normal stress on SE particles during lithiation for different SOC (a). Histogram of the normal stresses (b).

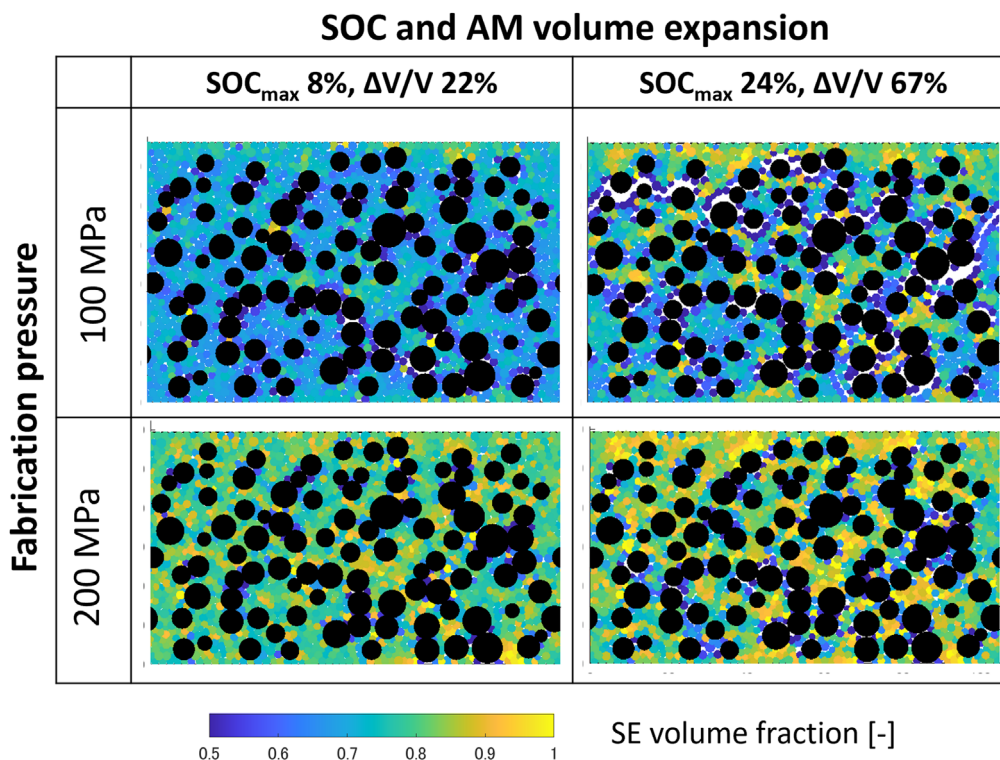


Figure 7. Electrode structures for different fabrication pressures after 5 intercalation cycles with different SOC.

the same time, ϕ_{SE} only varied slightly within 10% during electrode expansion which suggests that the change in κ_{SE}^{rel} is dominated by change in τ_{SE} . The large variation in τ_{SE} can be explained by the development of cracks, as observed in Fig. 7, that greatly increases the conductivity path. In Fig. 9b the time series of κ_{SE}^{rel} are plotted with different fabrication pressures. Although periodic variations can be seen, the permanent decrease in κ_{SE}^{rel} is significantly less when the fabrication pressure is higher.

Contact area.—Figure 10 show time series of the contact area, between AM and SE, obtained from Eq. 9. AM-SE contact area is plotted with the cycle number for different SOC_{max} in Fig. 10a. The contact area is higher during lithiation and increases with SOC_{max}. The contact area during delithiation decrease between cycles, especially at the first cycle. The time series of contact area is shown in Fig. 10b with different fabrication pressures. The fabrication pressure not only increases the overall contact area, but also reduces the contact area losses with time.

Discussion

Novelty and importance of this study.—A new DEM model with ductile particles was developed that was able to simulate cold press fabrication and mechanical degradation of ASSBs. The implementation of plasticity in the DEM model is novel, not only in the context of battery electrodes but also in the general context of particle simulation methods. With a two-step simulation scheme of cold pressing and intercalation, the effect of the fabrication conditions such as pressure on the electrode durability could be simulated. This is different from many earlier studies where the mechanical degradation in most cases has been simulated with a predetermined structure. From our results it was observed that intercalation induced expansion cause buildup of large stresses which could lead to formation of cracks inside. Electrode fabrication under a higher pressure was less susceptible to internal pore and crack formation.

With the new model it is also possible in the future to investigate the effect of material properties such as hardness on the mechanical degradation. Thus, the model developed can become a valuable tool to optimize fabrication processes and for selection adequate materials.

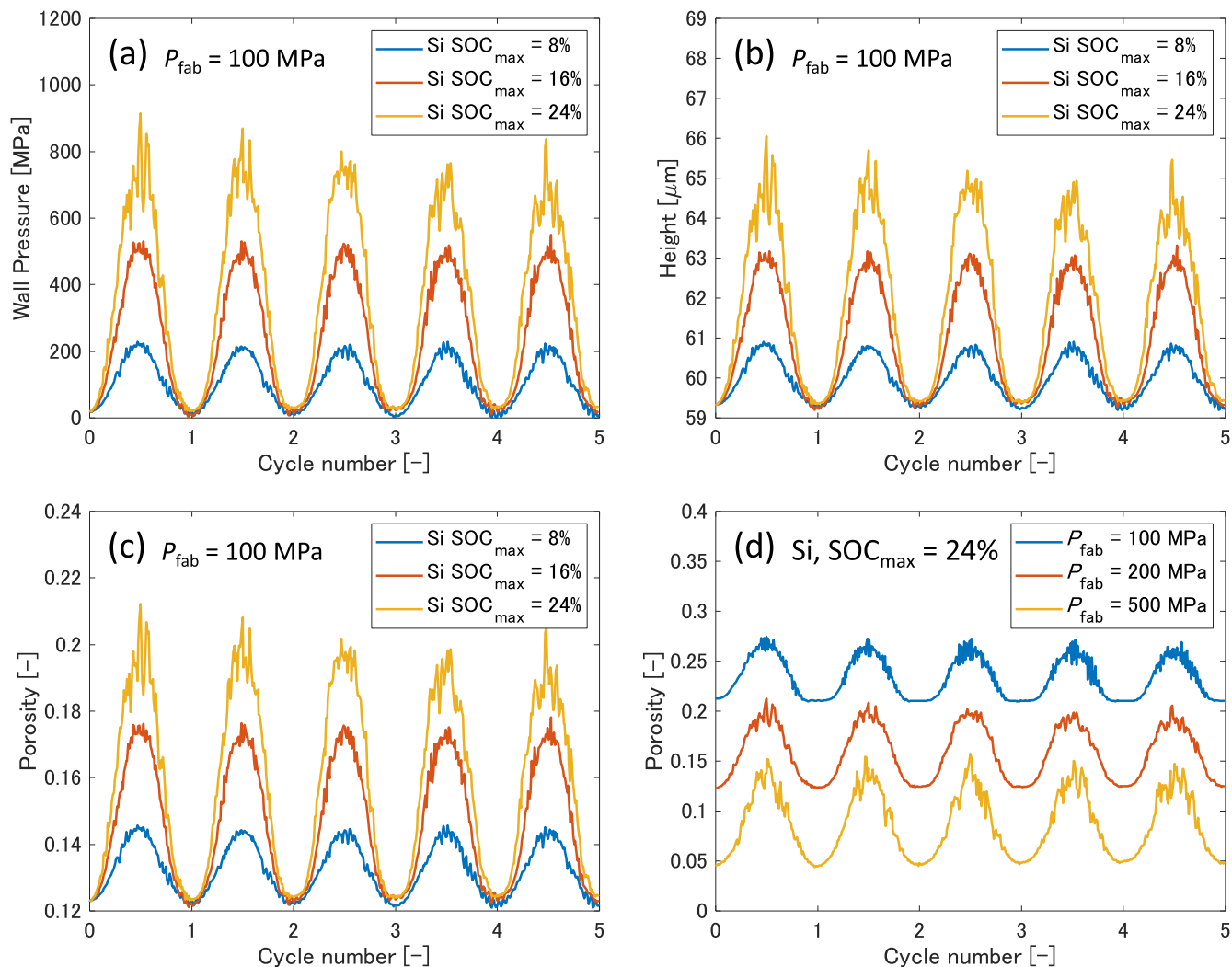


Figure 8. Time series of height (a), SE volume fraction (b) and porosity (c) for different charging conditions. Time series of porosity for different fabrication pressures (d).

Shortcomings of the study.—In this study, we simulated only in two dimensions. In Fig. 4 in Results, it was demonstrated that a 2D cluster DEM model may overestimate the solid fraction at low fabrication pressure. This may also affect the contact area between the particles. Even though our simulations were conducted in 2D, we assumed the domain to have a thickness in the z-direction for the purpose of contact area estimation and volume fractions. The shape of the particles was assumed to be spherical which simplified the calculations considerably. We included the effect of sub-particle contact indirectly by assigning a non-dimensional contact area ratio, α . In the AM-SE simulations, we limited the number of particles to only 1600. For higher accuracy this number should increase to assure particle number independence.

In the intercalation expansion simulation, we use a spring force calculated from the elasticity of the SE layer and the cathode layer. We did not include contraction of the foil layers. We did not either include foam packing which can be used when there is intercalation of high expansive materials such as Si. The latter can cause some of the pressure values to go beyond practical range of battery operation.

Although we calculated the tortuosity and the relative conduction coefficient of SE during cyclic loading, we did not calculate the coupled reaction and mass transport. We assumed the AM particle to expand in the same rate throughout the anode layer without consideration of the reaction distribution inside the electrode and AM. In the case of high discharge rate, the AM particles near the SE layer may expand more than those further away. This would also

affect the distribution of stress, contact area, cracking and tortuosity in the electrode.

Future research.—In order to improve the accuracy of our model, the natural step for improvement is to extend the simulation to 3 dimensions. We are currently working on non-spherical particles which can be simulated in DEM by the multisphere method. We have already simulated this type of DEM in previous studies for fuel cells catalyst layers.³²

For future validation of our model, we are planning to compare our results with micromechanical compression tests and/or nanoindentation tests. By varying the rate of change of the applied force during nanoindentation; elastic and plastic effects can be distinguished. Plastic deformation can be observed by hysteresis curves from the nano-indentation tests.

By coupling the DEM simulation with reaction and mass transfer, the distribution of AM expansion can be calculated through the reaction distribution. Because the particles are expanding, such calculation have to deal with moving interfaces which can be solved by phase field reaction model.³⁷ The plasticity model can be also applied to simulate dendrite growth with a phase field model and simulate the evolution of cracks.^{38,39} Such a model can simulate the progression of dendrite and the interaction with cracks inside the electrode layer.

Although we limited the study for ductile SE materials, the model can be extended to simulate brittle materials with some modifications.

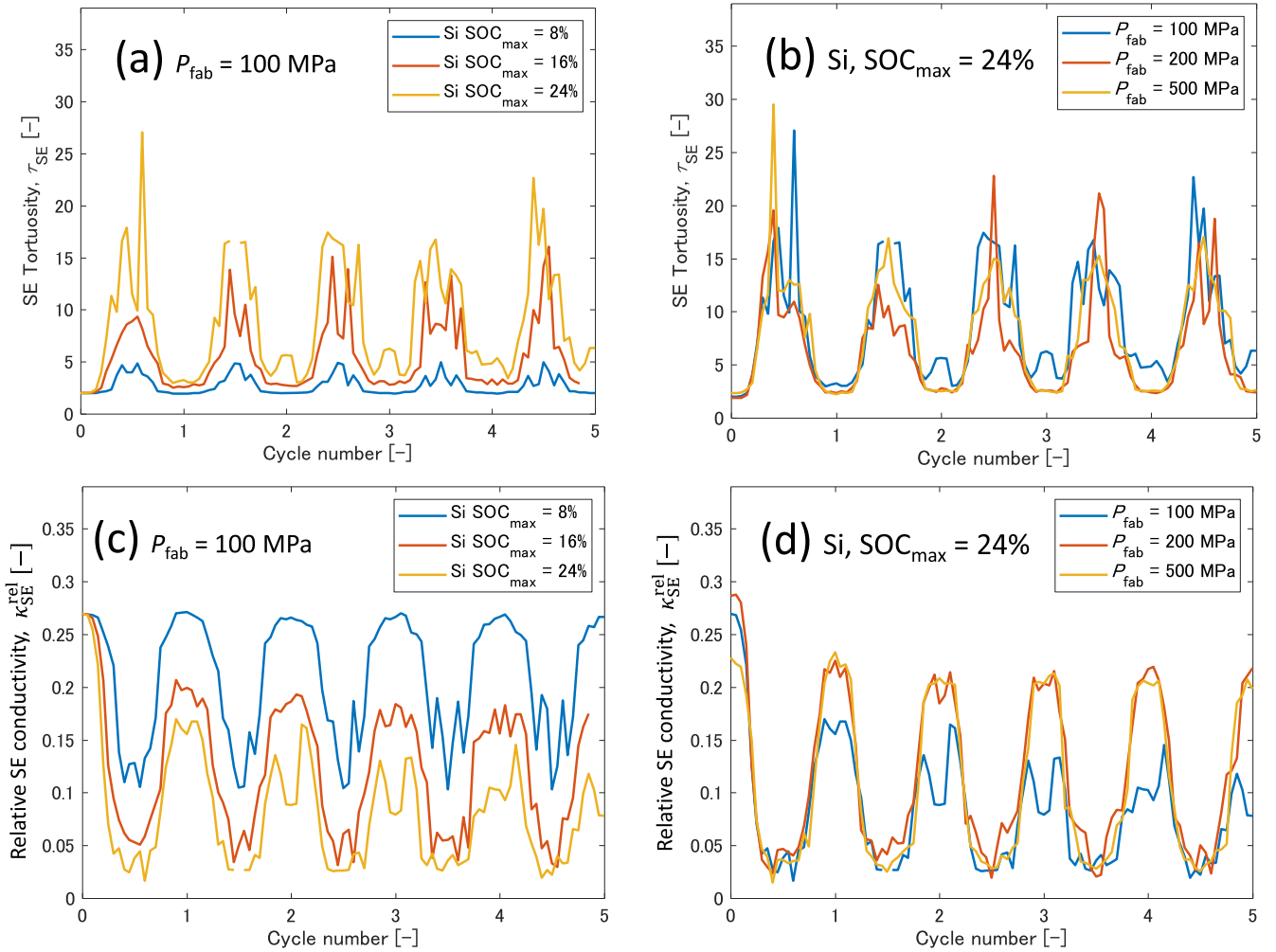


Figure 9. Time series of tortuosity with different charging conditions (a) and different fabrication pressures (b). Similarly, time series of relative SE conductivity (c), (d).

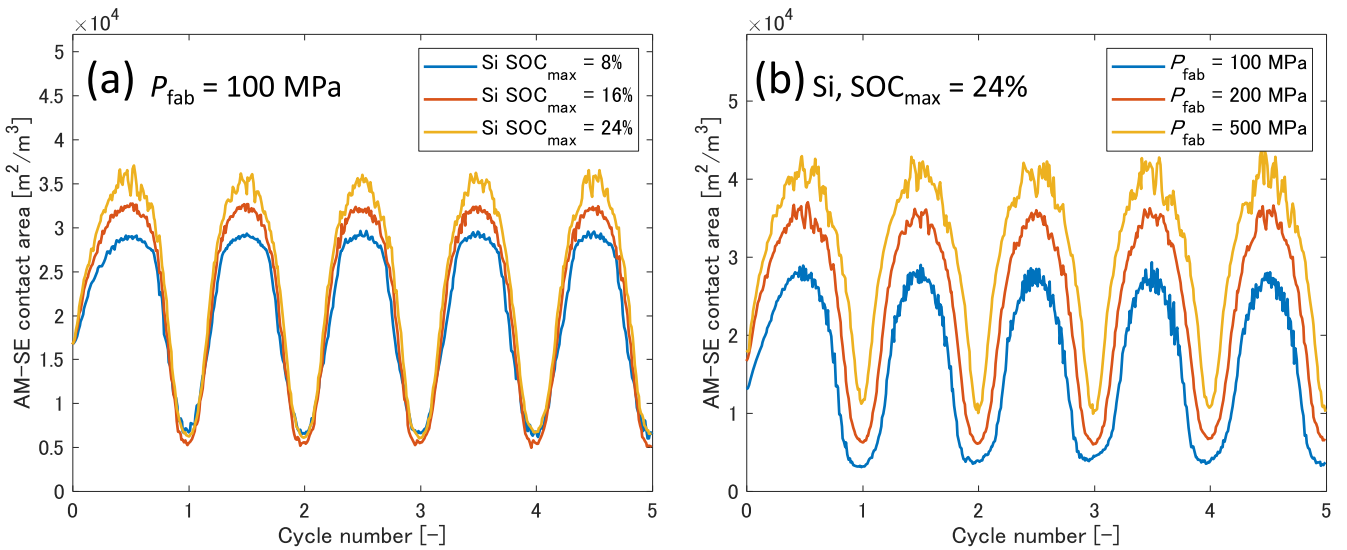


Figure 10. Time series of AM-SE contact area for different charging conditions (a) and different fabrication pressures (b).

Firstly, the tensile strength of such materials is usually much weaker than the compressive, so it is necessary to simulate with different yield strength depending whether the particles are approaching or separating

from each other. This model can also be applied for other fields such as powder packing fabrication and granular jamming control, where plastic deformation between particles is important.

Conclusions

The interaction during cold press and intercalation involves complex particle interactions including elastic and plastic deformation. The novelty of this study is the development of a new plasticity model, that can simulate the transition from granular media to solid electrode product during cold pressing; and utilize it for simulation of crack propagation during cyclic charge operation. This was achieved by using DEM with particles that with time becomes interconnected with cohesive beams when they are subjected to high pressure. In the case of low fabrication pressure, the particles become only partially consolidated and regions with voids remains within the electrode which weakens the internal structure. When the fabrication pressure is high, significant plastic deformation occur with the SE material that strengthens the cohesive bonds and minimize the voids.

The fabrication pressure had a large impact on the electrode damage during intercalation. The main reasons are attributed to local voids which cause appearances of local crack nucleation sites in the structure. The model developed is best suited for ductile SE such as dilithium sulfate where plastic deformation during cold pressing have been confirmed.^{9,10} However, with some modification the model can be extended to simulate brittle materials. The model opens up new research topics for the future such as growth and deformation of dendrite and how it can penetrate through cracks during intercalation expansion and contraction.

The model helps in the understanding of the mechanical processes during fabrication and operation. The ability of both being able to simulate fabrication and intercalation, makes it possible to directly relate fabrication conditions, such as the fabrication pressure, to the presence of voids and the effect of crack formation. This model can simulate the effect of material properties on the electrode durability and can therefore aid in the selection of suitable electrode materials in future ASSBs. In summary, this model is a valuable tool to aid in the electrode design to improve the durability and the performance of future All-Solid-State Batteries.

Acknowledgments

This work was supported by JSPS KAKENHI Grant-in-Aid for Scientific Research on Innovative Areas, “Science on Interfacial Ion Dynamics for Solid State Ionics Devices, Computational & Data Science (Gp-A03)”, grant number 19H05815; and by MEXT “Program for Promoting Researches on the Supercomputer Fugaku” (Fugaku battery & Fuel Cell Project), grant number JPMXP1020200301.

ORCID

Magnus So  <https://orcid.org/0000-0002-5358-3110>
Gen Inoue  <https://orcid.org/0000-0003-2427-0972>

References

- W. Li, J. R. Dahn, and D. S. Wainwright, *Science*, **264**, 1115 (1994).
- C. Sun, J. Liu, Y. Gong, D. P. Wilkinson, and J. Zhang, *Nano Energy*, **33**, 363 (2017).
- R. A. Huggins, *J. Power Sources*, **81–82**, 13 (1999).
- D. Ma, Z. Cao, and A. Hu, *Nano-Micro Lett.*, **6**, 347 (2014).
- H. Kim, E.-J. Lee, and Y.-K. Sun, *Mater. Today*, **17**, 285 (2014).
- R. Miyazaki, *Front. Energy Res.*, **8** (2020).
- F. Zheng, M. Kotobuki, S. Song, M. O. Lai, and L. Lu, *J. Power Sources*, **389**, 198 (2018).
- X. Ke, Y. Wang, G. Ren, and C. Yuan, *Energy Storage Mater.*, **26**, 313 (2020).
- A. Sakuda, A. Hayashi, and M. Tatsumisago, *Sci. Rep.*, **3**, 1 (2013).
- A. Kato, M. Nose, M. Yamamoto, A. Sakuda, A. Hayashi, and M. Tatsumisago, *J. Ceram. Soc. Japan*, **126**, 719 (2018).
- S. D. Fabre, D. Guy-Bouyssou, P. Bouillon, F. L. Cras, and C. Delacourt, *J. Electrochem. Soc.*, **159**, A104 (2011).
- G. Inoue and M. Kawase, *J. Power Sources*, **342**, 476 (2017).
- F. Itoh, G. Inoue, and M. Kawase, *ECS Trans.*, **69**, 83 (2015).
- H. Fathiannasab, A. G. Kashkooli, T. Li, L. Zhu, and Z. Chen, *J. Electrochem. Soc.*, **167**, 100558 (2020).
- A. Bielefeld, D. A. Weber, and J. Janek, *J. Phys. Chem. C*, **123**, 1626 (2019).
- A. Bielefeld, D. A. Weber, and J. Janek, *ACS Appl. Mater. Interfaces*, **12**, 12821 (2020).
- P. A. Cundall and O. D. L. Strack, *Géotechnique*, **29**, 47 (1979).
- C. S. Giménez, L. Helmers, C. Schilde, A. Diener, and A. Kwade, *Chem. Eng. Technol.*, **43**, 819 (2020).
- T. Shi, Q. Tu, Y. Tian, Y. Xiao, L. J. Miara, O. Kononova, and G. Ceder, *Adv. Energy Mater.*, **10**, 1902881 (2020).
- B. Störakers, S. Biwa, and P.-L. Larsson, *Int. J. Solids Struct.*, **34**, 3061 (1997).
- W. Leclerc, H. Haddad, and M. Guessasma, *Int. J. Solids Struct.*, **108**, 98 (2017).
- J. Tomas, *KONA Powder and Particle Journal*, **18**, 157 (2000).
- S. Luding, *Granular Matter*, **10**, 235 (2008).
- G. Bucci, T. Swamy, Y.-M. Chiang, and W. C. Carter, *J. Mater. Chem. A*, **5**, 19422 (2017).
- G. Bucci, B. Talamini, A. Renuka Balakrishna, Y.-M. Chiang, and W. C. Carter, *Phys. Rev. Materials*, **2**, 105407 (2018).
- C. V. Di Leo, E. Rejovitzky, and L. Anand, *Int. J. Solids Struct.*, **67–68**, 283 (2015).
- X. Gao, P. He, J. Ren, and J. Xu, *Energy Storage Mater.*, **18**, 23 (2019).
- Y. Liu, K. Guo, C. Wang, J. Han, and H. Gao, *Int. J. Solids Struct.*, **198**, 99 (2020).
- H.-K. Tian and Y. Qi, *J. Electrochem. Soc.*, **164**, E3512 (2017).
- B. N. J. Persson, *Surf. Sci. Rep.*, **61**, 201 (2006).
- F. P. McGrogan, T. Swamy, S. R. Bishop, E. Eggleton, L. Porz, X. Chen, Y.-M. Chiang, and K. J. V. Vliet, *Adv. Energy Mater.*, **7**, 1602011 (2017).
- M. So, K. Park, T. Ohnishi, M. Ono, Y. Tsuge, and G. Inoue, *Int. J. Hydrogen Energy*, **44**, 32170 (2019).
- M. So, K. Park, Y. Tsuge, and G. Inoue, *J. Electrochem. Soc.*, **167**, 013544 (2020).
- V. A. Sethuraman, M. J. Chon, M. Shimshak, N. Van Winkle, and P. R. Guduru, *Electrochem. Commun.*, **12**, 1614 (2010).
- M. N. Obrovac, L. Christensen, D. B. Le, and J. R. Dahn, *J. Electrochem. Soc.*, **154**, A849 (2007).
- S. J. Cooper, A. Bertei, P. R. Shearing, J. A. Kilner, and N. P. Brandon, *SoftwareX*, **5**, 203 (2016).
- L. Liang, Y. Qi, F. Xue, S. Bhattacharya, S. J. Harris, and L.-Q. Chen, *Phys. Rev. E*, **86**, 051609 (2012).
- C. Miehe, F. Welschinger, and M. Hofacker, *J. Mech. Phys. Solids*, **58**, 1716 (2010).
- X. Zhang, A. Krischok, and C. Linder, *Comput. Meth. Appl. Mech. Eng.*, **312**, 51 (2016).